# Supplementary Material

## Case implementation and provenance for the six-gate audit

This supplement accompanies “A Benchmark-Integrity Audit Protocol for Conformal Battery State-of-Health Evaluation.” It makes the retrospective case implementation auditable. The integrated G1–G6 synthesis was developed after the diagnostic program; it was not prospectively registered. The component plan files, scripts, and current paper data are preserved at immutable public replication commit 42f1120ed1f5cbc0282a706ebd4bec24a8d19d6e.

### Table S1. Analysis-to-gate implementation map

| ID | Governed analysis | Gate(s) | Tasks and rows | Feature schema | Model / conformal family |
|---|---|---|---|---|---|
| A01 / exp001 | Pooled-domain, cell-held-out coverage baseline | G1, G2, G3 | Main; governed cell-disjoint train/ calibration/validation/test partitions | QA numeric schema; recent prev_soh permitted only as a declared sequential measurement | Ridge, gradient boosting (GB), random forest, and extra trees with standard, weighted, domain-Mondrian, online-adaptive, and persistence-anchor conformal candidates; selected diagnostic is persistence-anchor protocol-Mondrian |
| A02 / exp002 | Recent-label and resistance ablation | G1 | Main; identical governed partitions across four conditions | Full; minus prev_soh; minus resistance; minus both; persistence candidate disabled when prev_soh is absent | Same candidate families as A01; validation-only selection within each condition |
| A03 / exp003 | Cross-dataset transfer diagnostic | G1, G2, G3 | LCAL, TCAL, LNAS, TNAS, LOxf, TOxf | Cross-domain numeric schema excludes dataset-, chemistry-, and protocol-identity columns; recent label remains a declared condition | Same point-predictor and conformal candidate families as A01, including protocol-Mondrian where groups exist |
| A04 / exp004 | Residual-bias stress screen | Ancillary to G3/G4 interpretation | Main test rows from A01 | No refit; additive bias is applied to the selected point prediction | Selected A01 persistence-anchor interval; conformal radius is fixed |
| A05 / exp005 | Host-timing screen | Ancillary only | 512 real Main test rows | prev_soh and precomputed calibration-group radius | Persistence-only lookup, persistence-anchor interval lookup, and GB inference; no deployment claim |
| A06 / exp006 | Method-family and dependence-aware comparison | G2, G3 | Main plus all six cross-dataset tasks | Governed task-specific schemas from the split manifest | Persistence-anchor protocol-Mondrian, GB quantile regression (QR), GB conformalized quantile regression (CQR), and NGBoost-CQR |
| A07 / exp007 | Cross-family UQ and two-threshold decision audit | G2, G3, G5 | Same seven tasks and corrected disjoint rows | Same task schemas as A06 | Persistence-anchor, GB-QR, GB-CQR, and NGBoost-CQR; decision cells at SOH 0.70 and 0.80 |
| A08 / exp008 | Reliability and decision-consequence audit | G1, G2, G3, G5 | Same seven tasks and corrected disjoint rows | Declared recent-label availability and task schemas | A07 families; cell-block uncertainty, subgroup checks, false acceptance, false rejection, and uncertainty |
| A09 / exp009 | Schema-cleaning and residual-shift audit | G4, G5 | Same seven tasks; added harmonized LNAS diagnostic | Original task schema versus a common cross-dataset schema | GB-CQR and supporting A08 families; identical rows within contrasts |
| A10 / exp010 | No-recent-label hard-regime audit | G1, G4, G5 | Seven tasks; horizons 0, 5, and 20; six SOH thresholds | No prev_soh; original, common-cycle, and stage schemas | GB-QR, GB-CQR, and stage-Mondrian CQR |
| A11 / exp011 | Schema-to-decision audit and independent control | G1, G4, G5 | Seven horizon-0 tasks; identical within-task rows; 245 contrast/end-point tests | As-used bundle, clean-original, common-cycle no-recent, common-cycle plus recent-label reference, and clean plus a fixed-seed permuted log-cycle control | One GB-CQR procedure fitted under each schema; paired cell-block deltas |
| A12 / exp012 | Localized-residual sensitivity comparator | G6 | Same seven corrected tasks and identical standard/localized rows | Clean task features shared by calibration and target rows; no density-ratio estimate | Standard GB-CQR versus Gaussian-kernel localized-residual CQR; sensitivity diagnostic only |

### Table S2. Locked parameters, selection, plan provenance, rerun, and deviations

| ID | Hyperparameters / search grid | Calibration and selection rule | Plan identifier, archive date, immutable plan commit | Governed run identifier | Recorded deviation or claim boundary |
|---|---|---|---|---|---|
| A01 | Seed 42; GB: 250 trees, depth 3, learning rate 0.035; RF/ET: 300 trees, minimum leaf 3; q grid 0.90, 0.93, 0.95, 0.97, 0.99, 1.00 | Validation coverage first, then minimum validation width; test metrics excluded | experiments/exp001_main/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545bdf528eb010 | PCR-2026-07-18-01 full real-data rerun; scripts/run_real_experiments.py --seed 42 | Legacy experiment lacks a standalone run manifest; current manuscript values were regenerated in the pinned full rerun and are covered by the paper SSOT/raw recomputation gate |
| A02 | Seed 42; four predeclared drop-feature conditions; A01 model and q grids | A01 validation-only rule, applied separately within each condition | experiments/exp002_ablation/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545bdf528eb010 | PCR-2026-07-18-01 full real-data rerun | The selected model family can change across conditions; therefore the Main contrast is pipeline sensitivity, not a method-matched feature effect |
| A03 | Seed 42; A01 model/q grids; target windows 0–20% train, 20–35% calibration, 35–55% validation, >55% test | Online candidate preferred only if validation coverage is at least 0.90; otherwise coverage-first/width selection | experiments/exp003_cross_protocol/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545b df528eb010 | PCR-2026-07-18-01 full real-data rerun | Negative transfer results were retained; no broad-transfer success claim |
| A04 | Bias grid −0.05, −0.02, −0.01, −0.005, 0, +0.005, +0.01, +0.02, +0.05 | No tuning or radius refit after bias injection | experiments/exp004_stress_failure/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545b df528eb010 | PCR-2026-07-18-01 full real-data rerun | Prediction-bias robustness screen only; not evidence of a physical degradation mechanism |
| A05 | 20 warmups, 300 timed repetitions, 512 real rows | Fixed timed path; no model selection | experiments/exp005_edge/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545bdf528eb010 | PCR-2026-07-18-01 full real-data rerun | Host amortized throughput only; explicitly excluded from edge/deployment evidence |

| ID | Hyperparameters / search grid | Calibration and selection rule | Plan identifier, archive date, immutable plan commit | Governed run identifier | Recorded deviation or claim boundary |
|---|---|---|---|---|---|
| A06 | Seed 42; 1,000 cell-block resamples; q grid 0.90–1.00; GB quantiles 0.05/0.95 with 250 trees, depth 3, learning rate 0.035; NGBoost 300 estimators, learning rate 0.03 | Validation coverage first, then width; contribution positioning fixed before result interpretation | experiments/exp006_fpa_repair/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912 e0051ddc23 | experiments/exp006_fpa_repair/results/ run_manifest.json; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | Closed as a diagnostic comparison rather than method superiority |
| A07 | Seed 42; 1,000 block resamples; q/GB/ NGBoost settings as A06; denominator floor 5 | Validation coverage-first width; inference intervals do not tune q | experiments/exp007_fpa_round2_repair/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912 e0051ddc23 | A07 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | July repair made target train/calibration/validation/ test windows pairwise disjoint and superseded earlier overlapping outputs |
| A08 | Seed 42; 1,000 block resamples; denominator floor 5; A07 method settings | Same-row method comparison; failed cells/tasks retained | experiments/exp008_reliability_audit/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db 141005f95b | A08 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | July repair preserves cluster multiplicity in cell resampling; thresholds remain illustrative decision cutoffs |
| A09 | Seed 42; 1,000 block resamples; denominator floor 5; q grid 0.90–1.00 | Identical rows and coverage-aligned comparisons; common-schema diagnostic added for LNAS | experiments/exp009_fpa_round4_repair/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db 141005f95b | A09 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | Common-schema cleaning cannot eliminate residual protocol drift; no physical-mechanism attribution |
| A10 | Seed 42; 1,000 block resamples; horizons 0/5/ 20; thresholds 0.65/0.70/0.75/0.80/0.85/0.90; denominator floor 5; q grid 0.90–1.00 | Validation-only q selection; Bonferroni and BH multiplicity summaries | experiments/exp010_hard_regime_audit/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db 141005f95b | A10 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | As-used no-recent schema rows containing bookkeeping fields are treated as contaminated and superseded by A11 contrasts |
| A11 | Seed 42; control-column seed 1033; 1,000 paired cell-block resamples; family size 5×7×7=245; family-wise alpha 0.05; denominator floor 5; GB settings as A06 | Identical task rows; Bonferroni-adjusted normal intervals from block-bootstrap SE; exact paired-cell sign test at nonzero zero-SE boundaries | experiments/exp011_original_paper_substance/ plan.md; 2026-06-01; ba5080bc38f9717aca59 19651b22db141005f95b | A11 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | The July repair replaced the earlier duplicated-fit control with an independently permuted log-cycle control. Because that single-draw control is non-null in LCAL/TCAL, it is a veto only and blocks mechanism attribution |
| A12 | Seed 42; 1,000 block resamples; denominator floor 5; bandwidth grid 0.25, 0.50, 1, 2, 4, infinity | Validation-only coverage-first width selection; test metrics excluded | experiments/exp012_shift_adaptive_cp_comparator/ plan.md; 2026-06-01; ba5080bc38f9717aca59 19651b22db141005f95b | A12 run manifest; corrected rerun 2026-07-18; code commit d934559297 9d2c929e611b54056b4176e79dbe15 | The implemented comparator is localized residual weighting, not a density-ratio or guarantee-bearing shift-adaptive conformal method; one point-coverage failure recovers, no lower-bound shortfall recovers |

## Provenance boundary

The plan-archive commits above precede the governed July 2026 rerun that supplies the current manuscript values. They establish that component analysis rules existed before the corrected rerun; they do not make the integrated six-gate synthesis prospectively registered. The earlier exploratory processed CSV could not be recovered byte-for-byte. The paper therefore reports only the corrected rerun regenerated from preserved raw holdings under Python 3.11.15, NumPy 2.2.6, pandas 2.3.3, and scikit-learn 1.7.2. Supplementary claims must be interpreted with the same case-bounded scope as the manuscript.